

- Requirements
 - Program must be able to allow the user to input a statement
 - Program must take the statement and count the number of specific type of coins and keep track of the obtained data
 - Program must account for base coin type and their plural forms
 - Program must account for capitalized letters
 - Based on the number of coins and their coin types, the program must calculate the total dollar amount
 - Output must be formatted in standard currency i.e X.XX

- Design(Pseudocode)

PRINT "Please enter your statement:"

READ user_input

SPLIT user_input INTO words

FOR each word IN words:

IF word is a numeric value THEN
 SET count TO integer value of word
 CONTINUE to next word
 ENDIF

IF word (lowercased) == "penny" OR word (lowercased) == "pennies" THEN
 SET cost TO 0.01
 ENDIF

IF word (lowercased) == "nickel" OR word (lowercased) == "nickels" THEN
 SET cost TO 0.05
 ENDIF

IF word (lowercased) == "dime" OR word (lowercased) == "dimes" THEN
 SET cost TO 0.10
 ENDIF

IF word (lowercased) == "quarter" OR word (lowercased) == "quarters" THEN
 SET cost TO 0.25
 ENDIF

ADD (count * cost) TO amount

SET count TO 0

END FOR

PRINT amount formatted to two decimal digits

END

- Implementation

- Source code can be found in Final_Sprint_2.py

- Testing

Test Cases:

- 1 penny and 2 nickels | Result: 0.11
- 1 PENNY AND 2 NICKELS | Result: 0.11
- 4 dimes and 7 quarters | Result: 2.15
- 4 DIMES AND 7 QUARTERS | Result: 2.15
- 1 quarter and 3 pennies | Result: 0.28
- 1 QUARTER AND 3 PENNIES | Result: 0.28
- 21 pennies and 17 dimes and 52 quarters | Result: 14.91
- 21 PENNIES AND 17 DIMES AND 52 QUARTERS | Result: 14.91
- 95 dimes and 73 quarters and 22 nickels and 36 pennies | Result: 29.21
- 95 DIMES AND 73 QUARTERS AND 22 NICKELS AND 36 PENNIES | Result: 29.21
- 1 nickel and 17 quarters | Result: 4.30
- 1 NICKEL AND 17 QUARTERS | Result: 4.30
- 21 nickels and 15 pennies | Result: 1.20
- 21 NICKELS AND 15 PENNIES | Result: 1.20
- 1 dime and 1 nickel and 1 penny and 1 quarter | Result: 0.41
- 1 DIME AND 1 NICKEL AND 1 PENNY AND 1 QUARTER | Result: 0.41
- 1 dime , 1 nickel . 1 penny , 1 quarter, Result: 0.41

The logic in this version of the program works as intended. There were no bugs or errors found during testing.